

Managing Uncertainty- Introduction

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Agenda

- Breaking the Ice
- Data Science?
- Managing Uncertainty
 - Syllabus
 - Course Policy
- Classification and Tabulation

Data Science

- Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to find unseen patterns, derive meaningful information, and make business decisions. Data science uses complex machine learning algorithms to build predictive models.
- The data used for analysis can come from many different sources and are presented in various formats.

Data Science Lifecycle

- **Capture:** Data Acquisition, Data Entry, Signal Reception, Data Extraction. This stage involves gathering raw structured and unstructured data.
- **Maintain:** Data Warehousing, Data Cleaning, Data Staging, Data Pre-processing, Data Architecture. This stage covers taking the raw data and putting it in a form that can be used.
- **Process:** Data Mining, Clustering/Classification, Data Modeling, Data Summarization. Data scientists take the prepared data and examine its patterns, ranges, and biases to determine how useful it will be in predictive analysis.
- **Analyze:** Exploratory/Confirmatory, Predictive Analysis, Regression, Text Mining, Qualitative Analysis. Here is the real meat of the lifecycle. This stage involves performing various analyses of the data.
- **Communicate:** Data Reporting, Data Visualization, Business Intelligence, Decision Making. In this final step, analysts prepare the analyses in easily readable forms such as charts, graphs, and reports.

- **Data acquisition** is the process of converting real-world signals to the digital domain for display, storage, and analysis. Because physical phenomena exist in the analog domain, i.e., the physical world that we live in, they must be first measured there and then converted to the digital domain.
- A **data warehouse** is a type of data management system that is designed to enable and support business intelligence (BI) activities, especially analytics.
- **Data staging** is the process of preparing and organizing data before it moves to its final destination, ensuring cleanliness and consistency for smooth integration. A data staging area provides the controlled space where this harmonization and preparation take place.
- **Data preprocessing** is an important step in data mining. Learn its basics, feature selection, feature engineering, data normalization, & more.
- **Data architecture** consist of models, policies, rules, and standards that govern which data is collected and how it is stored, arranged, integrated, and put to use in data systems and in organizations.

- Data are collections of any number of related observations.
- A collection of data is called a data set, and a single observation is a data point.
- Statisticians gather data from a sample. They use this information to make inferences about the population that the sample represents.
- Thus, a population is a whole, and a sample is a fraction or segment of that whole.

Managing Uncertainty

- **Uncertainty:**

- Data collected from sensors, social media, financial records, etc. is inherently uncertain due to noise, incompleteness, and inconsistency.
- Predictive analysis
- Application of probability theory

Classification and Tabulation

- The data collected for the purpose of a statistical inquiry sometimes consists of a few fairly simple figures, which can be easily understood without any special treatment.
- But more often there is an overwhelming mass of raw data without any structure. Thus, the unwieldy, unorganized and shapeless mass collected is not capable of being rapidly or easily associated or interpreted.
- Unorganized data are not fit for further analysis and interpretation.
- In order to make the data simple and easily understandable the first task is condensing and simplifying them.
- The procedure adopted for this purpose is known as the method of classification and tabulation. Classification helps with proper tabulation.

Classification

- Classification is a process of arranging things or data in groups or classes according to their resemblances and affinities and gives expressions to the unity of attributes that may subsist among a diversity of individuals.
- Classification is the process of arranging data into sequences and groups according to their common characteristics or separating them into different but related parts.

- Secrist
- The process of grouping large number of individual facts and observations on the basis of similarity among the items is called classification.

- Stockton & Clark

- **Characteristics of classification**

- Classification performs homogeneous grouping of data
- It brings out points of similarity and dissimilarities
- The classification may be either real or imaginary
- Classification is flexible to accommodate adjustments

- **Objectives / purposes of classifications**

- To simplify and condense the large data
- To present the facts easily in an understandable form
- To allow comparisons
- To help to draw valid inferences
- To relate the variables among the data
- To help further analysis
- To eliminate unwanted data
- To prepare a tabulation

- **Guiding principles (rules) of classifications**
- Following are the general guiding principles for good classifications:
 - Exhaustive: Classification should be exhaustive. Each and every item in the data must belong to one of the classes. Introduction of residual class (i.e. either, miscellaneous etc.) should be avoided.
 - Mutually exclusive: Each item should be placed in only one class.
 - Suitability: The classification should confirm to object of inquiry.
 - Stability: Only one principle must be maintained throughout the classification and analysis.
 - Homogeneity: The items included in each class must be homogeneous.
 - Flexibility: A good classification should be flexible enough to accommodate new situations or changed situations.

Modes / Types of Classification

- Modes / Types of classification refer to the class categories into which the data could be sorted out and tabulated. These categories depend on the nature of data and the purpose for which data is being sought.
 - Geographical (i.e. on the basis of area or region wise)
 - Chronological (On the basis of Temporal / Historical, i.e. with respect to time)
 - Qualitative (on the basis of character/attributes)
 - Numerical, quantitative (on the basis of magnitude)

Geographical Classification

- This kind of classification is based on geographical or location differences based between the various items like states, cities, regions, zones, areas etc.
- Geographical classifications are generally listed in alphabetical order or listed by the frequency size to emphasize the importance of various geographical regions.
- The following table shows the geographical distribution

States	Yield of Food-grains in (kg/acre)
Uttarakhand	36.8
Uttar Pradesh	36.9
Punjab	36.6
Haryana	34.6
Bihar	34.5
Jharkhand	35.2
Maharashtra	34.2

Chronological Classification

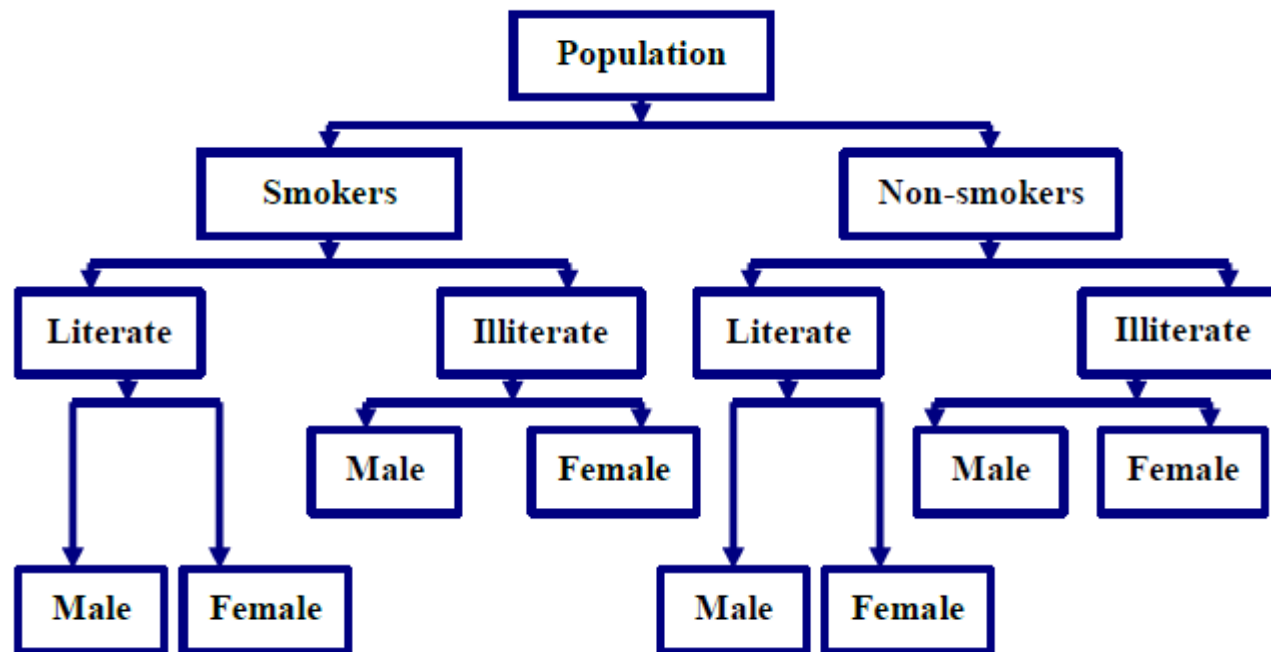
- In the chronological classification data are observed over a period of time.
- Such series are also known as time series because one of the variables in them is time.
- Time series are usually listed in chronological order, normally starting with the earliest period.
- The following table would give an idea of chronological classification.

Year	Birth Rate
2005	36.8
2006	36.9
2007	36.6
2008	34.6
2009	34.5
2010	35.2
2011	34.2

Qualitative Classification

- In qualitative classifications, the data are classified according to the presence or absence of attributes in given units. Thus, the classification is based on some quality characteristics/attributes.
- Ex: Gender, Literacy, Education, Class grade etc.
- Further, it may be classified as
 - Simple classification
 - Manifold classification
- Simple classification: If the classification is done into only two classes then classification is known as simple classification.
- Ex:
 - Population into Male / Female
 - Population into Educated / Uneducated

- Manifold classification: In this classification, the classification is based on more than one attribute at a time.



Quantitative Classification

- In Quantitative classification, the classification is based on quantitative measurements of some characteristics, such as age, marks, income, production, sales etc.
- The quantitative phenomenon under study is known as variable and hence this classification is also called classification by variable.
- For a 50 marks test, Marks obtained by students as classified as follows:

Marks	No. of students
00-10	5
11-20	7
21-30	10
31-40	25
41-50	3

Tabulation

- Tabulation may be defined, as a systematic arrangement of data in columns and rows.
- It is designed to simplify the presentation of data for the purpose of analysis and statistical inferences.
- The purpose of a table is to simplify the presentation and to facilitate comparisons.

- **Major Objectives of Tabulation**

- To simplify the complex data
- To facilitate comparison
- To economize the space
- To draw valid inference/conclusions
- It locates omissions and errors in the data
- To help for further analysis

- **Basic Principles of Tabulation**

- Tables should be clear, concise & adequately titled.
- Every table should be distinctly numbered for easy reference.
- Column headings & row headings of the table should be clear & brief.
- Units of measurement should be specified at appropriate places.
- Explanatory footnotes concerning the table should be placed in appropriate places.
- The Source of information of data should be clearly indicated.
- The columns & rows should be clearly separated with dark lines.
- Demarcation should also be made between data of one class and that of another.
- Comparable data should be put side by side.
- The figures in percentage should be approximated before tabulation.
- The figures, symbols etc. should be properly aligned and adequately spaced to enhance the readability of the same.
- Abbreviations should be avoided.

- **Differences between Classification and Tabulation**

- First data are classified and presented in tables; classification is the basis for tabulation.
- Tabulation is a mechanical function of classification because in tabulation classified data are placed in rows and columns.
- Classification is a process of statistical analysis while tabulation is a process of presenting data as a suitable structure.